RESEARCH DEPARTMENT - ANNUAL REPORT 2024

Prepared by: Dr. Olivia Chambers, Research Director

EXECUTIVE SUMMARY

This report summarizes the research activities and achievements of the department for fiscal year 2024. Our team of 45 researchers across three divisions has produced 23 peer-reviewed publications.

TEAM LEADS AND KEY PERSONNEL

Computational Biology Division:
Lead: Dr. Benjamin Foster
Senior Researchers: Dr. Priya Sharma, Lucas Zimmermann
Notable: Published groundbreaking work on protein folding

Machine Learning Division:
Lead: Dr. Fatima Al-Rashidi
Senior Researchers: Christopher Wong, Anna Kowalski
Notable: Developed novel transformer architecture

Applied Mathematics Division:
Lead: Prof. Michael O'Sullivan
Senior Researchers: Dr. Elena Volkov, Raj Krishnamurthy
Notable: Breakthrough in optimization algorithms

EXTERNAL COLLABORATIONS

Our department maintained active collaborations with several institutions. Dr. Priya Sharma co-authored a paper with Prof. Hans Weber from ETH Zurich.

Dr. Fatima Al-Rashidi presented at NeurIPS alongside Dr. James Chen from Stanford University.

AWARDS AND RECOGNITION

- Dr. Benjamin Foster received the Young Investigator Award from the National Science Foundation.
- Anna Kowalski was named to the Forbes 30 Under 30 list in the Science category.
- Christopher Wong received a best paper award at ICML.

BUDGET AND RESOURCES

Total research budget: $4.2M

External grants secured: $1.8M

- NIH Grant (PI: Dr. Benjamin Foster): $600K
- NSF Grant (PI: Dr. Fatima Al-Rashidi): $450K
- DARPA Contract (PI: Prof. Michael O'Sullivan): $750K

LOOKING AHEAD

For 2025, we plan to expand the ML division under Dr. Fatima Al-Rashidi. New hires will include two postdoctoral researchers and a visiting scholar.

Submitted by:
Dr. Olivia Chambers
Director of Research

Reviewed by: Margaret Thompson, CEO
Date: December 15, 2024